

CTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$ (100k)	0.316	1.11	0.625	0.38
$W_1 \downarrow$ (200k)	0.292	0.659	0.457	0.331
$W_1 \downarrow$ (300k)	0.287	0.512	0.384	0.322
$W_1 \downarrow$ (400k)	0.283	0.451	0.351	0.317
$W_1 \downarrow$ (500k)	0.283	0.407	0.35	0.314
Cramér $\ell_2^2 \downarrow$ (100k)	0.0379	0.133	0.089	0.0427
Cramér $\ell_2^2 \downarrow$ (200k)	0.0301	0.0486	0.0513	0.0312
Cramér $\ell_2^2 \downarrow$ (300k)	0.0284	0.0308	0.0411	0.0295
Cramér $\ell_2^2 \downarrow$ (400k)	0.0275	0.0249	0.0373	0.0288
Cramér $\ell_2^2 \downarrow$ (500k)	0.0271	0.0215	0.0372	0.0282
CF- $\ell_2^2/\omega^2 \downarrow$ (100k)	0.0314	0.105	0.0759	0.0354
CF- $\ell_2^2/\omega^2 \downarrow$ (200k)	0.024	0.0388	0.044	0.0254
CF- $\ell_2^2/\omega^2 \downarrow$ (300k)	0.0223	0.025	0.0345	0.0236
CF- $\ell_2^2/\omega^2 \downarrow$ (400k)	0.0212	0.0201	0.0316	0.0229
CF- $\ell_2^2/\omega^2 \downarrow$ (500k)	0.0211	0.0171	0.0309	0.0222

QTD

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$ (100k)	0.243	0.913	0.573	0.299
$W_1 \downarrow$ (200k)	0.221	0.557	0.423	0.257
$W_1 \downarrow$ (300k)	0.201	0.464	0.403	0.251
$W_1 \downarrow$ (400k)	0.206	0.407	0.382	0.252
$W_1 \downarrow$ (500k)	0.2	0.398	0.365	0.247
Cramér $\ell_2^2 \downarrow$ (100k)	0.018	0.103	0.0805	0.0225
Cramér $\ell_2^2 \downarrow$ (200k)	0.013	0.0355	0.0484	0.0145
Cramér $\ell_2^2 \downarrow$ (300k)	0.0101	0.0236	0.0463	0.0138
Cramér $\ell_2^2 \downarrow$ (400k)	0.0103	0.0173	0.0421	0.0135
Cramér $\ell_2^2 \downarrow$ (500k)	0.00958	0.017	0.0388	0.0136
CF- $\ell_2^2/\omega^2 \downarrow$ (100k)	0.0173	0.0865	0.0765	0.0214
CF- $\ell_2^2/\omega^2 \downarrow$ (200k)	0.0126	0.0317	0.0473	0.0138
CF- $\ell_2^2/\omega^2 \downarrow$ (300k)	0.00975	0.0223	0.0458	0.0132
CF- $\ell_2^2/\omega^2 \downarrow$ (400k)	0.01	0.0165	0.0417	0.013
CF- $\ell_2^2/\omega^2 \downarrow$ (500k)	0.00925	0.0165	0.0385	0.0131

ϕ TD Dirac

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$ (100k)	0.154	0.606	0.297	0.177
$W_1 \downarrow$ (200k)	0.143	0.483	0.251	0.179
$W_1 \downarrow$ (300k)	0.152	0.439	0.251	0.167
$W_1 \downarrow$ (400k)	0.136	0.452	0.231	0.179
$W_1 \downarrow$ (500k)	0.132	0.432	0.23	0.155
Cramér $\ell_2^2 \downarrow$ (100k)	0.0067	0.0412	0.0171	0.00557
Cramér $\ell_2^2 \downarrow$ (200k)	0.00554	0.0184	0.0123	0.00554
Cramér $\ell_2^2 \downarrow$ (300k)	0.00665	0.0154	0.0136	0.00466
Cramér $\ell_2^2 \downarrow$ (400k)	0.00509	0.017	0.011	0.0055
Cramér $\ell_2^2 \downarrow$ (500k)	0.00464	0.0139	0.0103	0.00377
CF- $\ell_2^2/\omega^2 \downarrow$ (100k)	0.00582	0.0334	0.0149	0.00474
CF- $\ell_2^2/\omega^2 \downarrow$ (200k)	0.0048	0.0153	0.0111	0.00484
CF- $\ell_2^2/\omega^2 \downarrow$ (300k)	0.00579	0.0129	0.0124	0.00382
CF- $\ell_2^2/\omega^2 \downarrow$ (400k)	0.00441	0.0151	0.00987	0.00461
CF- $\ell_2^2/\omega^2 \downarrow$ (500k)	0.00393	0.0117	0.00917	0.00305

ϕ TD Cat

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$ (100k)	0.295	0.798	0.47	0.328
$W_1 \downarrow$ (200k)	0.29	0.494	0.403	0.318
$W_1 \downarrow$ (300k)	0.292	0.447	0.372	0.318
$W_1 \downarrow$ (400k)	0.289	0.423	0.366	0.319
$W_1 \downarrow$ (500k)	0.289	0.422	0.347	0.317
Cramér $\ell_2^2 \downarrow$ (100k)	0.0299	0.0834	0.0515	0.03
Cramér $\ell_2^2 \downarrow$ (200k)	0.0281	0.0314	0.0394	0.0289
Cramér $\ell_2^2 \downarrow$ (300k)	0.0289	0.0245	0.0365	0.0286
Cramér $\ell_2^2 \downarrow$ (400k)	0.0281	0.0225	0.0363	0.029
Cramér $\ell_2^2 \downarrow$ (500k)	0.0283	0.0224	0.0353	0.0281
CF- $\ell_2^2/\omega^2 \downarrow$ (100k)	0.024	0.0633	0.0443	0.0242
CF- $\ell_2^2/\omega^2 \downarrow$ (200k)	0.0223	0.026	0.0341	0.0228
CF- $\ell_2^2/\omega^2 \downarrow$ (300k)	0.0226	0.0207	0.0314	0.0228
CF- $\ell_2^2/\omega^2 \downarrow$ (400k)	0.022	0.0197	0.031	0.0231
CF- $\ell_2^2/\omega^2 \downarrow$ (500k)	0.0221	0.0184	0.0299	0.0226

ϕ TD Quant

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$ (100k)	0.161	0.377	0.198	0.175
$W_1 \downarrow$ (200k)	0.152	0.352	0.204	0.167
$W_1 \downarrow$ (300k)	0.151	0.371	0.189	0.157
$W_1 \downarrow$ (400k)	0.137	0.389	0.207	0.163
$W_1 \downarrow$ (500k)	0.144	0.383	0.189	0.167
Cramér $\ell_2^2 \downarrow$ (100k)	0.00694	0.0121	0.00971	0.00511
Cramér $\ell_2^2 \downarrow$ (200k)	0.00573	0.00932	0.00898	0.00433
Cramér $\ell_2^2 \downarrow$ (300k)	0.00603	0.00943	0.00782	0.00373
Cramér $\ell_2^2 \downarrow$ (400k)	0.0046	0.0107	0.00899	0.00449
Cramér $\ell_2^2 \downarrow$ (500k)	0.00525	0.0102	0.00862	0.00432
CF- $\ell_2^2/\omega^2 \downarrow$ (100k)	0.00628	0.0113	0.00923	0.00443
CF- $\ell_2^2/\omega^2 \downarrow$ (200k)	0.00511	0.00882	0.00851	0.00375
CF- $\ell_2^2/\omega^2 \downarrow$ (300k)	0.00539	0.00858	0.00733	0.0031
CF- $\ell_2^2/\omega^2 \downarrow$ (400k)	0.00406	0.00975	0.00847	0.004
CF- $\ell_2^2/\omega^2 \downarrow$ (500k)	0.00462	0.00935	0.00813	0.00351